A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 01

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 4 points below. **1** Sigmoidal (S-shaped) growth curve **1** Slow start (few breeding individuals), then rapid / exponential growth (years 4-10) as abundant food and space mean birth rate far exceeds death rate **1** Growth slows as density-dependent factors (intra-species competition for food, disease) raise the death rate **1** Population stabilises / fluctuates around about 920-950 - the carrying capacity - where birth rate equals death rate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carrying capacity** → the largest population the area can support **Do not credit** • 'the population cannot grow any more' as a definition - numbers fluctuate around the carrying capacity	4

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 2 points below. **1 Density-dependent:** effect on survival increases with population density - food shortage / intra-species competition, disease spread, predation **1 Density-independent:** chance of survival unaffected by density - drought, flood, volcanic eruption, severe winter **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Without wolves (natural predators removed) the population is regulated only by starvation / disease at a high density **1** At this density deer over-graze, preventing tree regeneration and damaging habitats / other species – culling keeps numbers below the damaging level (as in Scotland) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Rising prey numbers give predators more food so predator numbers rise – high predation then reduces the prey population **1** Food shortage reduces predators, allowing prey to recover - cyclic fluctuations with predators lagging behind prey **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 78-81; AQA spec 3.1.3.4.	
02.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Its mass	1
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** The 24-hour rotation period means the surface is not exposed to sunlight for long, so temperature extremes / excessive heating and cooling are reduced **1** The axis is tilted relative to the orbit so the amount of insolation at each latitude changes through the year, producing seasonal variations in conditions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **insolation** → incoming solar radiation / energy arriving from the Sun	2

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** The distance from the Sun gives surface temperatures between 0 and 35 °C in most areas: too close and water would boil / evaporate, too far and it would freeze **1** Atmospheric pressure (produced by the mass of gases held by gravity) is high enough to stop the water boiling / evaporating away – at much lower pressure there would be no liquid water **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	
03.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Environmental DNA (eDNA) analysis	1
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Transmitters are too heavy for small birds / their batteries would not last for a long-term study **1** Geolocator tags are small and only record time and light levels – latitude and longitude for each day are worked out when the bird is recaptured and the data downloaded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Each separate (regional) population has its own unique gene pool **1** DNA from seized ivory can be compared with the database to identify where the elephant came from, so poaching hotspots / trade routes can be targeted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-405; AQA spec 3.1.2.3.6.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 3 points below. **1 Hard release:** individuals released with no post-release support - suitable for species whose behaviour is instinctive and need no learned skills (insects, fish, amphibians, reptiles) **1 Soft release:** gradual release into larger areas with food provided while they learn to find food, avoid predators - needed for mammals and birds **1 Problems faced:** finding / recognising food and poisonous foods, hunting skills (captive-born predators never learned), avoiding predators, acceptance into wild social groups **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3
04.2	**Any 5 of 7** 1 mark each for any 5 of the 7 points below, in any order, maximum 5. **1** Large enough suitable habitat for a viable population **1** Reliable food supply **1** Low predation risk / original threat removed **1** Suitable breeding sites **1** Water **1** Support of the local human population **1** Official / legal protection of habitat and species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Reliable** → repeatable / gives consistent results when repeated **Do not credit** • Points after the first five - the question asks for five, so a list of extras cannot pick up the marks. • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	5
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Tracking (radio / GPS tags, rings) shows survival, dispersal and breeding of released animals so the success of the project can be assessed **1** Identifies causes of failure so future releases can be improved / extra support given **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 36-37; AQA spec 3.1.2.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** They are driven by renewable, low energy-density resources, especially solar energy, which will not be depleted: the hydrological cycle, the carbon and nitrogen cycles, atmospheric and thermohaline circulation **1** Most natural processes occur at low temperatures, because enzymes lower the activation energy of the reactions, so little energy is needed **1** Processes link into cycles in which the waste product of one is the raw material for the next, so raw materials are not depleted and wastes do not build up **1** A small number of abundant, simple elements are built into monomers and then into a wide range of polymers, so supplies of the raw materials are secure **1** Wastes are biodegradable and break down to non-toxic products, so they do not accumulate or poison the system **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **thermohaline** → the ocean conveyor driven by temperature and salinity • **biodegradable** → broken down by decomposers **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Minerals are dispersed after use - spread thinly through products all over the world, or lost to corrosion - so there is no concentrated deposit left to recover **1** Products are made of mixtures of materials that are difficult to separate, so recovery and re-use are expensive and often incomplete **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2

Do not write outside the box

Question	Marking guidance	Marks
05.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1 Design for extended use:** a product with a long life, which can be repaired, gives fewer opportunities for its materials to be lost **1 Design for end-of-life re-use:** materials that can be identified and separated easily are far more likely to be recycled **1** Separation of technical and biological materials, because metals and plastics need different reprocessing from biodegradable materials, which can be composted or digested **1** Energy from renewable sources at every stage, as in natural systems **1** Optimum rather than maximum production, so that the system supports the processes it relies on instead of maximising one output **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	3
05.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Sustainability depends on the total impact, which is the number of people multiplied by the impact of each person, so the same activity crosses the threshold as either rises **1** A natural process can absorb or replace what is taken only up to a rate – beyond that the resource is depleted or the waste accumulates faster than it can be broken down **1 Example:** slash and burn agriculture is sustainable where the population is small enough for each clearing to be left fallow long enough to recover, but not when land has to be re-cleared before the soil fertility has returned **1 Example:** discharging untreated sewage into a river is harmless where decomposers can break it down, but causes deoxygenation and eutrophication once the load exceeds that rate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **threshold** → the concentration below which no harm is detected • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3

Do not write outside the box

Question	Marking guidance	Marks
05.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Urban populations are supplied by resources brought in from elsewhere, so the damage caused by extraction happens out of sight and often in another country **1** In a sedentary rural community the resource is local, so the problems of over-exploitation become obvious quickly as supplies dwindle, and a cultural belief in limiting exploitation to needs develops **1 Consequence:** when one area is exhausted, exploitation simply moves to another area and the consumer sees no change, so the total damage keeps growing - this has happened with timber, energy, metal minerals and fish **1 Consequence:** there is no pressure from consumers to change, so unsustainable practices continue until the resource fails on a large scale **1 Consequence:** the costs fall on the people and ecosystems where extraction happens, not on those who benefit, so the market price never reflects the true environmental cost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-377; AQA spec 3.6.2-3.6.3.	
06.1	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Hedgerows **1** Hay meadows **1** Grazed moorland / heathland **1** Chalk grassland **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Drainage produces more aerobic soils so wetland plants and the animals they support cannot survive **1** Fertilisers increase growth of tall species such as grasses which out-compete smaller plants such as wildflowers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2

Do not write outside the box

Question	Marking guidance	Marks
06.3	1 mark for the point below. **1** The introduced (non-indigenous) predator may eat other species as well as the pest **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biological control** → using a natural predator, parasite or pathogen of the pest	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-329; AQA spec 3.5.1.3.	
07.1	**1 mark each** 1 mark for each of the 4 points below. **1 SSSI:** UK legislation (Wildlife and Countryside Act) – best examples of UK flora, fauna, geological features – management plan agreed with owner who must notify Operations Likely to Damage (ploughing, drainage, pesticides, tree planting / removal) **1 SPA:** EU Birds Directive – places of international importance for birds (with SACs under the Habitats Directive form the Natura 2000 network; may restrict activities in surrounding areas) **1 Ramsar:** Ramsar Convention (intergovernmental agreement) – internationally important wetlands, originally for waterfowl, now wetland conservation and wise use generally **1 MCZ:** UK marine legislation (Marine and Coastal Access Act) under the W&CA framework – protects marine habitats / species by restricting activities such as fishing and dredging – NNR: best SSSIs with complete communities **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	4
07.2	**1 mark each** 1 mark for each of the 3 points below. **1 Legal status:** restrictions on damaging activities within the area (and sometimes in surrounding areas, eg drainage) **1** Management agreements between owners and statutory bodies to maintain suitable conditions **1** But it does not protect against external influences such as air pollution or climate change – very few sites are state-owned so protection depends on owner cooperation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
07.3	**1 mark each** 1 mark for each of the 3 points below. **1** Much of the UK's wildlife lives in habitats created by farming (hay meadows, hedgerows) which only survive if the farming continues **1** Managing land for wildlife costs money / reduces income so farmers need payments (points for beetle banks, hedgerows, buffer strips, wild bird seed, low-input grassland) **1** Benefits everyone so society pays through central funding – legislation cannot compel positive management **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30, 38-39; AQA spec 3.1.2.3.2.	
08.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1 Subsistence maize:** 9.0 / 0.6 = 15 **1 Intensive wheat:** 54.0 / 18.0 = 3 **1 Feedlot beef:** 12.0 / 40.0 = 0.3 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. **Do not credit** • A bare wrong answer with no working.	3
08.2	**1 mark each** 1 mark for each of the 3 points below. **1** An energy subsidy is the energy put into a farming system by people rather than by the Sun: fuel for machinery, the manufacture of fertilisers and pesticides, irrigation, transport, drying and storage **1** Intensive systems raise the yield per hectare by adding large energy subsidies, so the output rises but the input rises faster and the ratio falls **1** The feedlot ratio is below 1, so it takes more energy to produce the beef than the beef contains – the extra trophic level means most of the energy in the feed is lost in respiration, excretion and the inedible parts of the animal **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **trophic level** → the feeding level in a food chain **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3

Do not write outside the box

Question	Marking guidance	Marks
08.3	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Keep the food chain short by producing crops for direct human consumption rather than feeding them to livestock, so one transfer of about 10% efficiency is avoided **1** Restrict the movement of livestock, so less of the energy in the feed is used for muscular activity and more is converted to biomass **1** Keep the animals warm, or rear them indoors, so less energy is spent maintaining body temperature **1** Slaughter the animals young, while their growth rate and food conversion ratio are still high **1** Use species or breeds with a high food conversion ratio, such as poultry or farmed fish, rather than cattle **1** Control parasites and disease, and use feed that is easy to digest, so more of the food eaten is assimilated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **food conversion ratio** → the feed needed per unit of food produced **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** A single desired gene can be transferred, so the characteristic appears without the unwanted genes that come with a whole parent in conventional breeding **1** Genes can be moved between species that could never interbreed, so characteristics can be introduced that selective breeding could never produce, and the result is achieved in one generation instead of many **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
08.5	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Genes may escape by pollen transfer to wild relatives or to neighbouring non-GM crops, creating herbicide-resistant weeds or making organic certification impossible **1** Insect-resistant crops may kill non-target species, including the pollinators and natural pest predators that the wider ecosystem depends on **1** Pests and weeds evolve resistance to the toxin or the herbicide the crop is designed to be used with, so the advantage is temporary and use of the chemical increases **1** Reliance on a few patented GM varieties narrows the genetic diversity of the crop, making it more vulnerable to a new disease **1** Seed is patented and must be bought each year, which can put subsistence farmers in debt or make them dependent on one supplier **1** Long-term effects on human health are not fully known, and labelling may not let consumers choose **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **genetic diversity** → the range of alleles / genes present in a population **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'GM crops are unnatural' as an environmental impact - credit gene flow, resistance, effects on non-target species or reduced pesticide use	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2.	
09.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 600 + (90 + 150) - (220 + 80) = 540 **1** Stock fell by 60 so the catch exceeds the MSY / unsustainable **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (thousand tonnes) is already printed on the answer line. • **MSY** → the largest catch that can be taken without reducing the stock **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks. • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
09.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** A + G - M = 90 + 150 - 80 = 160 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (thousand tonnes) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
09.3	**1 mark each** 1 mark for each of the 3 points below. **1** Catch is landed and weighed at ports so it is recorded **1** Other terms need data on fish in the sea: populations are mobile (crossing national boundaries), very unevenly distributed **1** Representative sampling of a large enough proportion of the area is impractical / expensive – survival of eggs and larvae varies greatly between years **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
09.4	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 CPUE:** if more effort (boat-days, engine power, net size) is needed to land the same catch, the population is declining even though total catch is maintained – a better measure than catch size which technology can prop up **2** Falling mean size / age (from scale or otolith growth rings) shows older adults are being removed faster than replaced and the catch depends on small young fish **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 349-351; AQA spec 3.5.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Context:** wild catches have not increased since about 1990 as stocks are fully or over-exploited – aquaculture output growing rapidly and approaching the wild catch – fish are the last large-scale hunted food • **Advantages of aquaculture:** control of limiting factors, selective breeding, gender control, high yields per area – fish have low FCRs (no thermoregulation / support) compared with livestock – herbivores (tilapia, carp) and molluscs very efficient – polyculture, IMTA, aquaponics, rice-fish • **Limits:** carnivorous species (salmon, shrimp) need fishmeal from wild fish so still depend on fishing and deplete forage fish (sandeels - puffins) – some species (bluefin tuna) will not breed in captivity – consumer preference for carnivores • **Environmental impacts:** habitat loss (mangroves), organic pollution, pesticides, antibiotics and resistance, lice, escapes and gene pool contamination, predator control – and how they can be reduced • Fishing can be sustainable with quotas, gear design, NTZs, size limits, effort limits – MSY management needs good data • **Judgement:** aquaculture can supply much of the growth in demand, especially herbivores and shellfish, but cannot fully replace fishing – both need sustainable management **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **aquaculture** → fish or shellfish farming	9
	• **gene pool** → the range of alleles / genes present in a population • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	
	Model answer - Level 3 (9 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Wild catches have not increased since about 1990 because most stocks are fully or over-exploited, while aquaculture has grown rapidly and now approaches the wild catch, so the question is how far it can take over. Its advantages are real: limiting factors, breeding, gender and disease can be controlled, yields per area are high, and fish have low food conversion ratios because they need not support their weight or keep warm; herbivores such as tilapia and carp and filter-feeding molluscs need no fishmeal at all.	
	Its limits are equally real. The carnivorous species consumers prefer - salmon, shrimp - are fed fishmeal from wild forage fish, so their farming still depends on fishing and depletes species such as sandeels on which puffins rely; some species, such as bluefin tuna, will not breed in captivity. Farms destroy mangroves, release organic waste, pesticides and antibiotics that breed resistance, spread lice, allow escapes that contaminate wild gene pools and provoke predator control, though siting, closed systems, polyculture and vaccines can reduce these. Meanwhile fishing itself can be sustainable if quotas based on good stock data, gear design, size and effort limits and no-take zones are enforced.	
	Aquaculture can therefore supply most of the growth in demand, especially through herbivorous fish and shellfish, and can relieve pressure on wild stocks, but it cannot fully replace fishing while carnivorous species depend on wild fishmeal and some species cannot be farmed. Both must be managed sustainably; neither is a substitute for the other.	
	Weaker answer - Level 1 (2 marks)	
	Aquaculture is farming fish instead of catching them. Fish farms can produce a lot of fish and they do not run out like wild fish do. But fish farms cause pollution and diseases can spread to wild fish. Salmon farms feed the salmon on fish which have to be caught. So aquaculture can replace some fishing but not all of it.	
	What makes the difference	
	• Level 1 reaches a reasonable conclusion but supports it with two sentences; Level 3 gives the context (flat wild catch since 1990), the advantages in detail and the impacts with ways to reduce them.	
	• Level 3 distinguishes between species types (herbivores and molluscs vs carnivores) - the key to 'to what extent'.	
	• Level 3 also considers whether fishing can be made sustainable, which the comparison requires.	
10.2	1 mark for the point below. **1** A species that is given artificial food (pellets) rather than relying on natural food in the water, eg salmon, shrimp **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-359; AQA spec 3.5.2.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Species selection:** market, range of tolerance, controllability • **Abiotic control:** temperature, light, water, soil fertility / nutrients, aeration, salinity, pH, CO_2, topography, relief, wind - methods and examples • **Biotic control:** pest groups, cultural control, biological control (successes and failures), sterile males, pheromones, genetic resistance, pesticides (properties), antibiotics, pollinators, soil biota • **Manipulation of food species:** population density, monocultures, selective breeding, asexual reproduction, cross-breeding, AI / embryo transfer, GM – energetics - intensive vs extensive, energy subsidies, ratios, FCR, food chain losses • **Impacts:** habitat loss / change, reduced biodiversity, introduced species, pesticide and nutrient pollution, GHGs, hydrological changes, soil erosion – positive: plagioclimax habitats • Social, economic and political influences shaping methods • Sustainable strategies and evaluation	25
	– conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 306-337; AQA spec 3.5.1.	

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Question	Marking guidance	Marks
11.2	**Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Marine productivity:** light, nutrients, upwelling – where fisheries occur • **Methods:** demersal / pelagic trawling, purse seining, drift nets, long lines, shellfish traps - effectiveness, selectivity, energy, impacts • **Impacts:** overfishing (r vs k selection, catchable vs breeding size), changed age structure, by-catch (albatross, porpoise, turtles, shrimp), ghost fishing, seabed / reef / seagrass damage, dynamite, food web effects (sandeels, tritons) • **Population dynamics and MSY:** Russell formula, difficulties of data, indicators from catches (CPUE, mean size, age), research on eggs and larvae • **Management:** quotas, mesh design, escape panels, TEDs, pingers, hook design, effort limits, method bans, NTZs, closed seasons, size limits, protected individuals, population seeding, biodegradable gear – EU CFP, IWC, CCAMLR • Evaluation with examples (cod, orange roughy, herring recovery, NTZ spillover) – role of aquaculture	25
	– judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-351; AQA spec 3.5.2.	

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